NEXUS SOP EVIDENCE

LOCAL RETRIEVAL

Source-grounded industrial knowledge assistant

Maintenance Session Report

RESOLVED

Automated maintenance and troubleshooting record | Phoenix Industrial Corp, Plant #4

Report ID REP-2026-8832

Session date	Time window	Total downtime	Final status
28 Oct 2026	11:15 - 11:42 AM	27 min	Resolved

1. Session overview and metadata

Machine
Robotic Packaging Arm
Line 2 | SKU #RPA-990

Operator
Rajesh Kumar
Badge #4092

Assisting tool
Multimodal GenAI Assistant
v2.1 Edge Edition

2. Incident and error diagnostic

Reported symptom
Arm stalling mid-cycle with erratic jittering.

Captured error code
ERR-909 Servo over-current / Axis Z obstruction

Multimodal input
Worker uploaded a smartphone photo of the Axis Z gearbox housing.

VLM visual analysis
The Vision-Language Model detected foreign debris buildup (plastic wrap jam) wedged in the drive gear teeth.

3. Retrieval and safety protocols

Referenced manual	**Robotic_Arm_Series_7_Maintenance_Guide_Rev4.pdf (pages 112-114)**
Wiring / schematic	**DWG-RPA-Z-Axis-Wiring.cad (grid section D-2)**

Safety interlock enforced: warning
Mandatory power disconnect and pneumatic pressure release required before clearing mechanical jams. Follow the approved site LOTO procedure and verify the zero-energy state before accessing the mechanism.

4. Step-by-step execution log

1. **Safety isolation** 11:17 AM
   Operator engaged emergency stop and toggled the LOTO safety switch at Main Panel B. Confirmed via smart-tablet sensor.
2. **Debris removal**
   Followed the AI prompt to remove safety guard #3 using a standard 5 mm Allen key, then extracted the jammed shrink-wrap film from the Axis Z drive teeth.
3. **Hardware inspection**
   Inspected gear teeth for stripping. Teeth integrity verified as nominal (100% intact).
4. **System reboot and test cycle**
   Cleared error code ERR-909 via the control pad and ran a diagnostic dry-run. Axis Z returned to normal calibration parameters.

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Troubleshooting flowchart | REP-2026-8832

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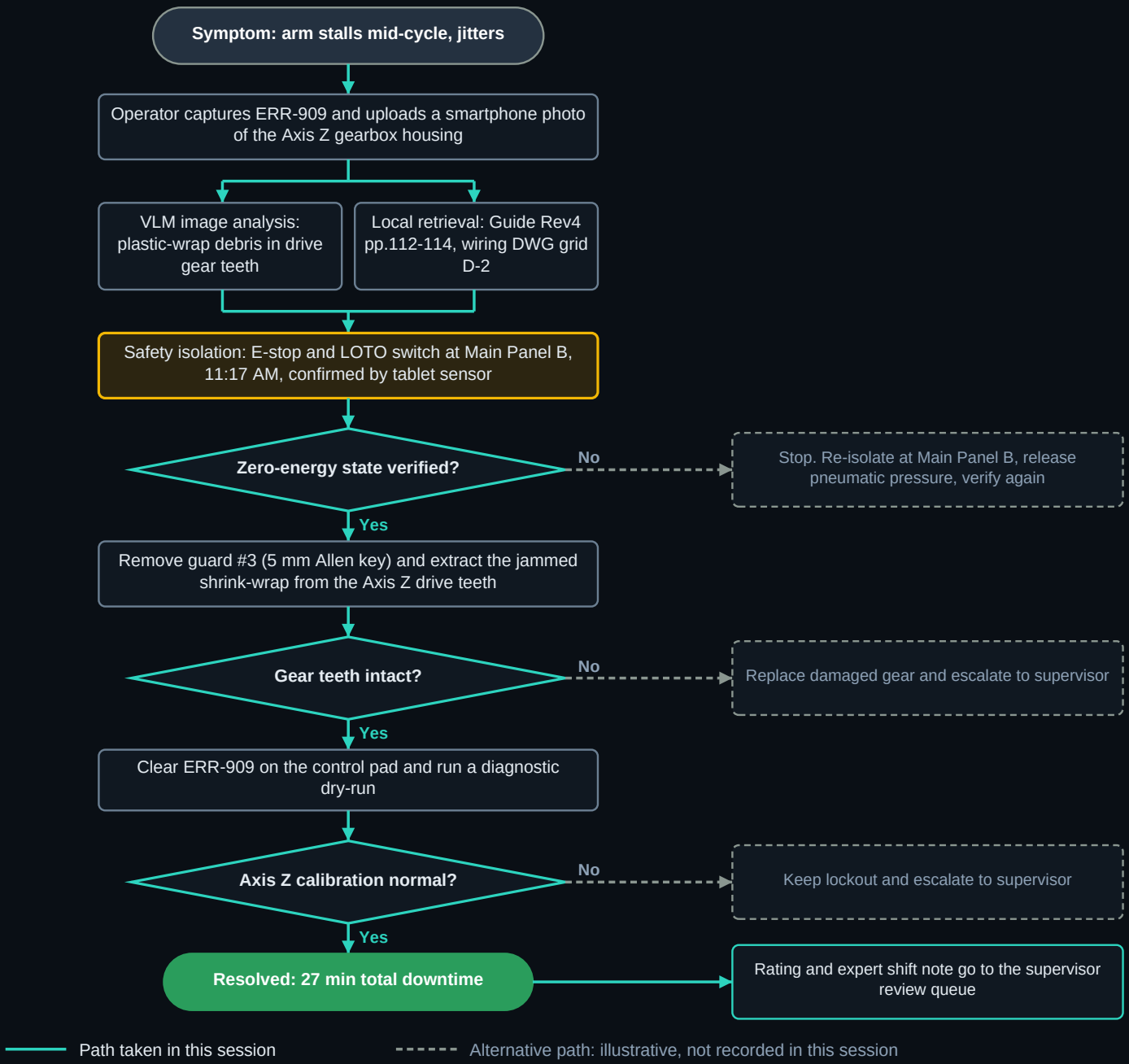
Figure 1. Troubleshooting flowchart for ERR-909, Axis Z



Maintenance Session Report

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Technical diagrams | Axis Z drive and fault chain

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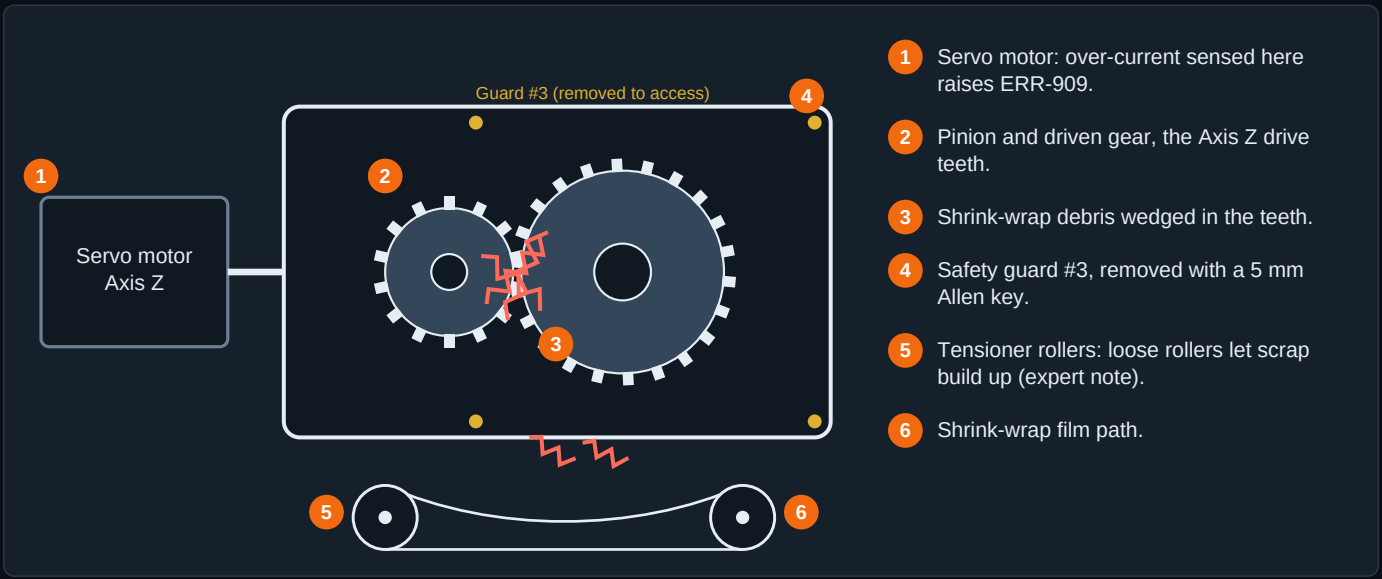
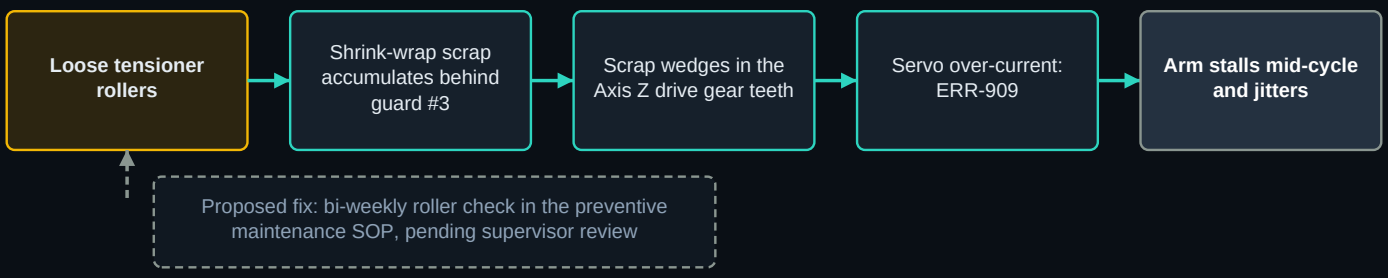
Figure 2. Axis Z drive schematic, showing where the jam occurred



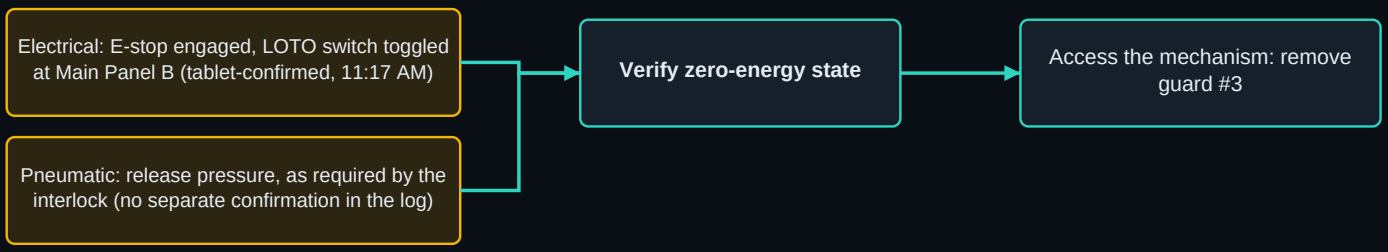
Illustrative schematic, not to scale and not taken from the CAD drawing. Gear and roller placement is simplified. For actual wiring see DWG-RPA-Z-Axis-Wiring.cad, grid section D-2.

Figure 3. Likely fault chain, from the operator's shift note



The chain is the operator's hypothesis. Plant Engineering has not yet confirmed it.

Figure 4. Energy isolation before touching the mechanism



Maintenance Session Report

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System architecture and feedback loop | REP-2026-8832

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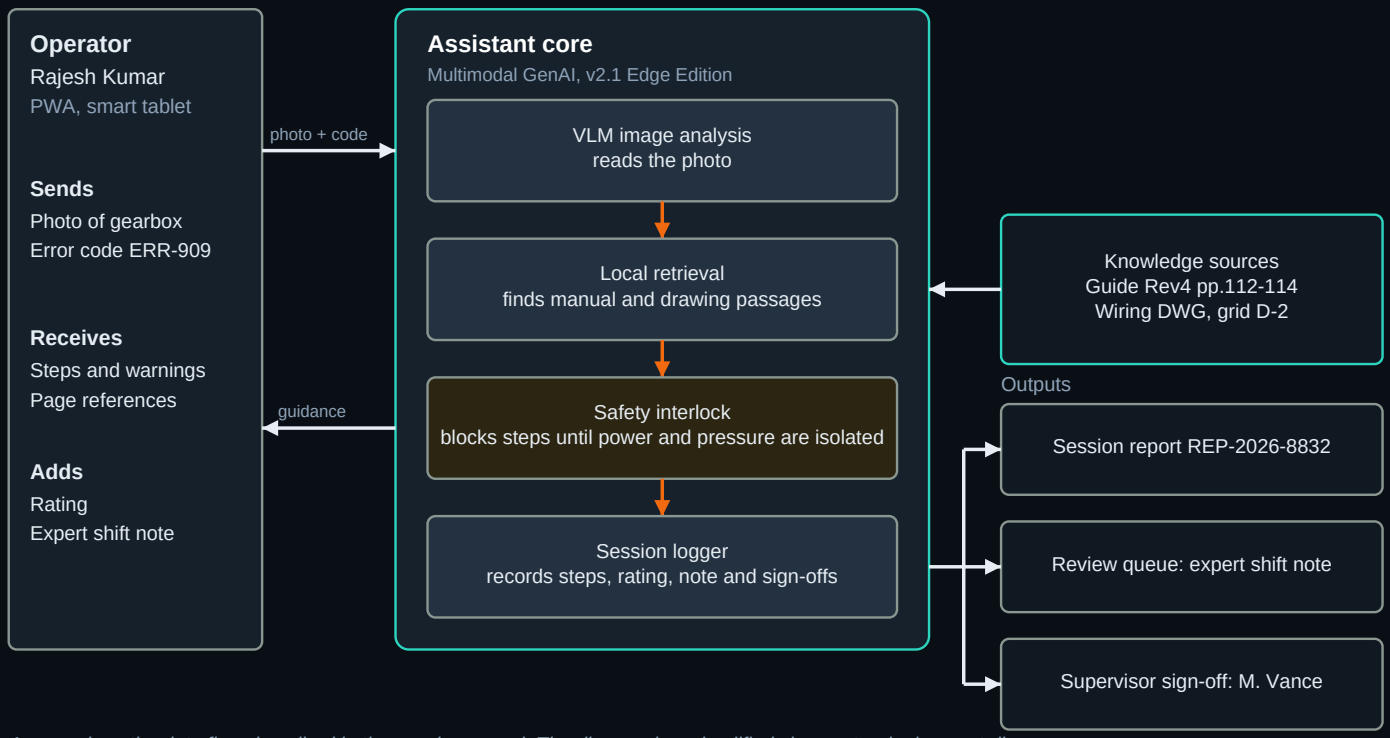
Figure 5. How the assistant produced this report



Arrows show the data flow described in the session record. The diagram is a simplified view, not a deployment diagram.

5. Worker feedback and shift log annotation

Operator rating

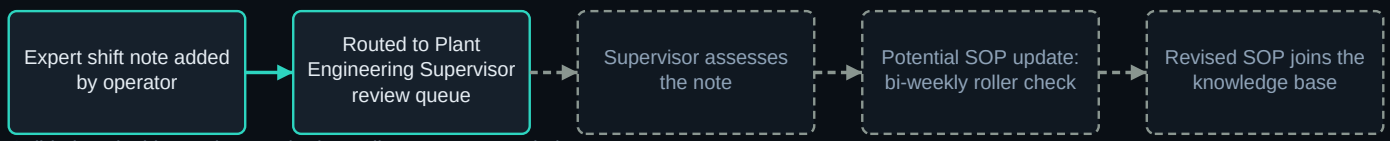
Highly accurate. Step-by-step guidance was clear.

> Expert shift note (added by operator)
>
> *"Shrink-wrap scrap tends to accumulate behind guard #3 if tensioner rollers are running loose. Recommend adding a bi-weekly roller check to preventive maintenance SOP."*

Knowledge base update

Note successfully routed and appended to the Plant Engineering Supervisor review queue.

Figure 6. Where the expert note goes next



Solid: done in this session. Dashed: pending, not yet recorded.

Maintenance Session Report

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Approval, audit trail and metadata | REP-2026-8832

Report ID REP-2026-8832

Outcome

Final session status	Total downtime	Operator rating
Resolved	27 min	Highly accurate

6. Approval and audit trail

Operator sign-off
Rajesh K.
Verified via PWA biometric / PIN
Verified

Supervisor sign-off
Maintenance Lead M. Vance
Auto-approved
Approved

Figure 7. Audit trail, in order

Statuses are shown as recorded in the session data.

Report metadata

Generated by	**Multimodal GenAI Industrial Assistant Core Engine**
Assistant version	**v2.1 Edge Edition**
Record type	**Plant engineering / maintenance**
Plant	**Phoenix Industrial Corp, Plant #4**
Machine	**RPA-990 / Line 2**
Report ID	**REP-2026-8832**

Record integrity note

This report reflects the session details supplied for this demonstration. Image findings, sensor confirmations, sign-offs and machine status are presented as reported and should be independently verified in the plant's approved maintenance system before being treated as an official operational record. Maintenance activities must follow current site procedures and qualified personnel authorization.

Generated automatically by Multimodal GenAI Industrial Assistant Core Engine.